DHCP CONFIGURATION – CISCO PACKET TRACER

Practical Assignment — Questions and Answers

1. Configure One DHCP Server and Two Client PCs

Question 1: In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

Answer

A Dynamic Host Configuration Protocol (DHCP) server automatically provides network settings to client devices. In this activity, one server will provide an IP address to each of two PCs. The clients should receive different addresses from the configured pool.

Network setup

1. Open Cisco Packet Tracer and place one Server, one Switch (for example, 2960) and two PCs (PC0 and PC1) in the workspace.
2. Connect Server, PC0 and PC1 to the switch using Copper Straight-Through cables.
3. Assign the server a static address. Open Server > Desktop > IP Configuration and enter IP address 192.168.1.2, subnet mask 255.255.255.0 and default gateway 192.168.1.1.

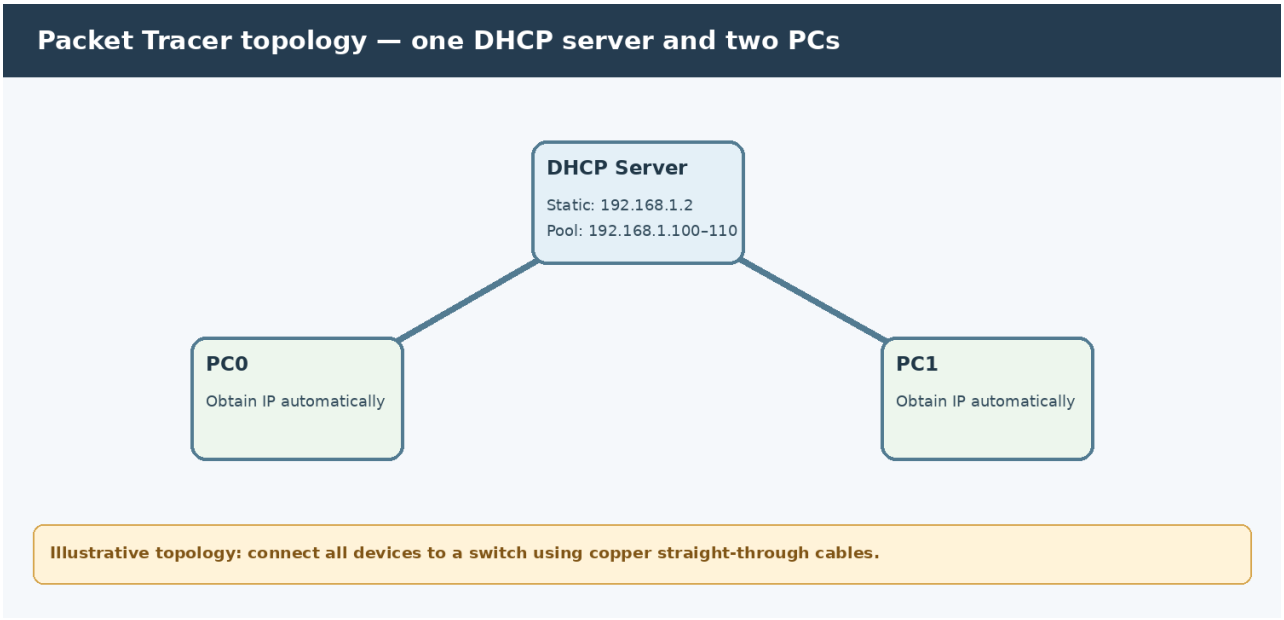


Figure 1. Suggested network layout. This is an illustrative topology, not a screenshot captured from Packet Tracer.

1. Configure One DHCP Server and Two Client PCs — Continued

Configure the DHCP service

4. Open Server > Services > DHCP and switch the DHCP service to On.
5. Create a pool named OFFICE-LAN.
6. Enter Default Gateway 192.168.1.1 and DNS Server 8.8.8.8.
7. Set Start IP Address to 192.168.1.100, Subnet Mask to 255.255.255.0 and Maximum Number of Users to 10. Click Add or Save, depending on the Packet Tracer version.
8. On PC0, open Desktop > IP Configuration and select DHCP. Repeat the same action on PC1.

Verify the assigned addresses

On each PC, open Desktop > Command Prompt and enter ipconfig. The PCs should receive different addresses from the pool, for example 192.168.1.100 and 192.168.1.101. The exact order may vary.

Device	Example IPv4 address	Subnet mask	Address source
PC0	192.168.1.100	255.255.255.0	DHCP pool
PC1	192.168.1.101	255.255.255.0	DHCP pool

Explanation and conclusion

The server leases an available address to each client. Since the pool contains multiple addresses, each PC can receive a unique IP address. The subnet mask identifies the local network range, while the gateway and DNS settings help clients communicate beyond the local network and resolve domain names.

Conclusion: DHCP simplifies network administration by automatically assigning unique IP addresses and related settings without manual configuration on every PC.

2. Draw the DHCP Message Sequence (DORA)

Question 2: Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage. Use arrows to show the direction of each message between client and server.

Answer

The DHCP address allocation process is commonly explained using the four-message sequence called DORA: Discover, Offer, Request and Acknowledgement (ACK). The client sends Discover and Request messages; the server sends Offer and ACK messages.

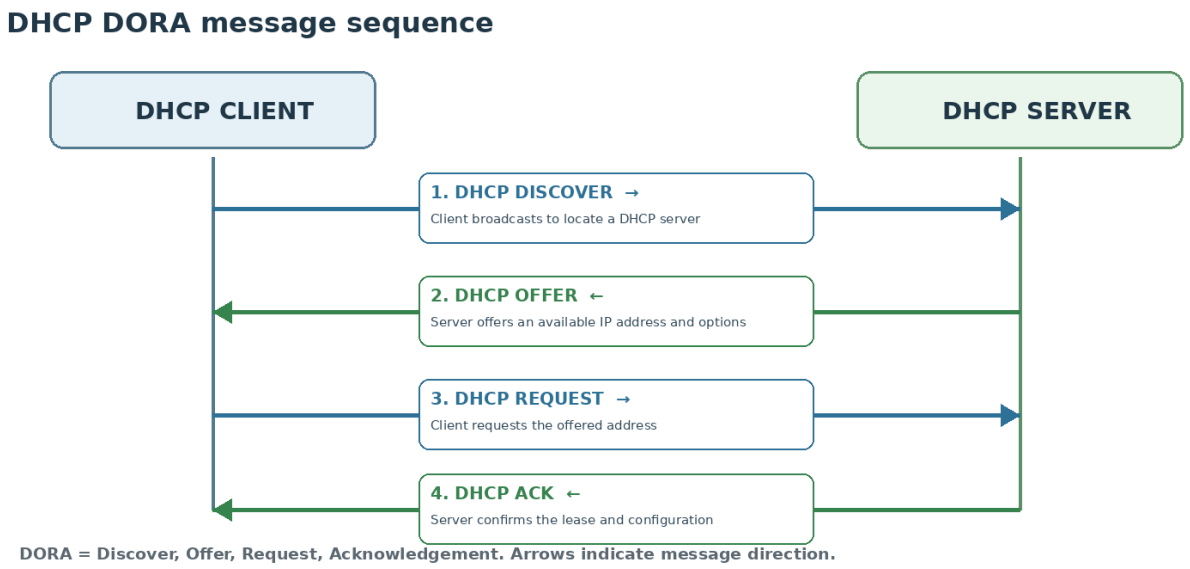


Figure 2. DHCP DORA sequence diagram. Horizontal arrows show which device sends each message.

Step	Sender → Receiver	Purpose
1. Discover	Client → DHCP Server	Client broadcasts to locate a DHCP server.
2. Offer	DHCP Server → Client	Server offers an available IP address and configuration options.
3. Request	Client → DHCP Server	Client requests the offered IP address.
4. Acknowledge (ACK)	DHCP Server → Client	Server confirms the lease and supplies network settings.

Important details

- The initial Discover message is generally broadcast because the client does not yet have a usable IP address or know the DHCP server.
- The Offer contains a proposed address and other configuration information.
- The Request indicates which offered address the client wants to use.
- The ACK confirms the lease. The client can then configure its network interface and communicate.

Conclusion: DORA describes the basic exchange that allows a client to obtain an IP address automatically from a DHCP server.

3. Test a DHCP Pool with Only One Available IP Address

Question 3: Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2–3 lines why only one PC succeeds.

Answer

In this experiment, the DHCP pool will contain only one usable address. Both PCs will request an address, but the server can lease the single available address to only one client at a time.

Procedure

9. Use the same server-and-switch topology with two PCs, or create a new simple topology.
10. Open Server > Services > DHCP. Edit the existing pool or remove it and create a new pool named SINGLE-IP.
11. Set the Start IP Address to 192.168.1.100, subnet mask to 255.255.255.0 and Maximum Number of Users to 1. Ensure that the DHCP service is On.
12. On both PCs, choose Desktop > IP Configuration > DHCP. If one PC already holds the lease, release/renew or toggle DHCP on the clients as needed to test the second request.

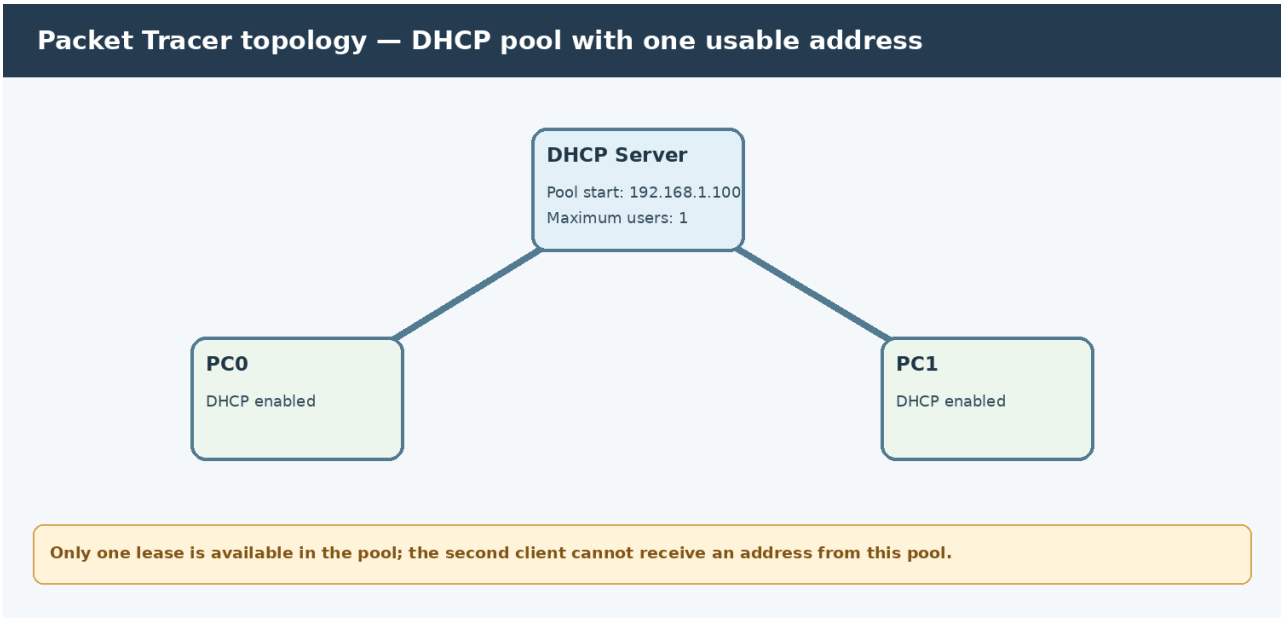


Figure 3. Topology for the one-address DHCP pool test. Illustrative layout only.

3. Test a DHCP Pool with Only One Available IP Address — Results

Expected observation

Illustrative result — one address in DHCP pool

Device	DHCP status	Assigned IPv4 address	Subnet mask
PC0	DHCP successful	192.168.1.100	255.255.255.0
PC1	DHCP failed / APIPA	No pool address	Not assigned

Which PC succeeds first can vary; the first client to obtain the only lease gets the address.

Figure 4. Illustrative result. Actual Packet Tracer output and which PC succeeds first may differ.

Example result: The first PC to obtain the lease receives 192.168.1.100. When the second PC requests an address, the pool has no free address remaining, so it cannot obtain a valid lease from that pool.

2–3 line explanation

Only one IP address is available in the DHCP pool, so the server can lease it to only one PC at a time. The second PC cannot receive an address from that pool because no free address remains; it may show a DHCP failure or an automatic private address, depending on the simulation.

What to include in your practical screenshot

- Show the DHCP pool settings with Start IP Address 192.168.1.100 and Maximum Number of Users set to 1.
- Show PC0 and PC1 IP Configuration or Command Prompt output so the successful and unsuccessful results are visible.
- If the Packet Tracer simulation does not display a clear failure message, show that one PC has the pool address and the other has no valid address from the configured pool.

Conclusion: A DHCP pool must contain enough available addresses for the devices expected to connect. A pool that is too small prevents additional clients from obtaining an address.

4. Simulate a Restaurant Guest Wi-Fi Network

Question 4: Simulate a Zomato-style restaurant Wi-Fi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.

Answer

A restaurant can provide guest Wi-Fi so customers can connect their devices and use online services such as food-ordering or restaurant apps. DHCP automatically assigns each connected device an IP address and network settings.

Suggested topology and configuration

Use a wireless access point or wireless router for the guest Wi-Fi network, a DHCP server, one laptop and one generic PC representing a smartphone. If the Packet Tracer version or device model does not support wireless association for the generic PC, use a wireless-capable end device or connect the representative PC to the access point using the available supported method.

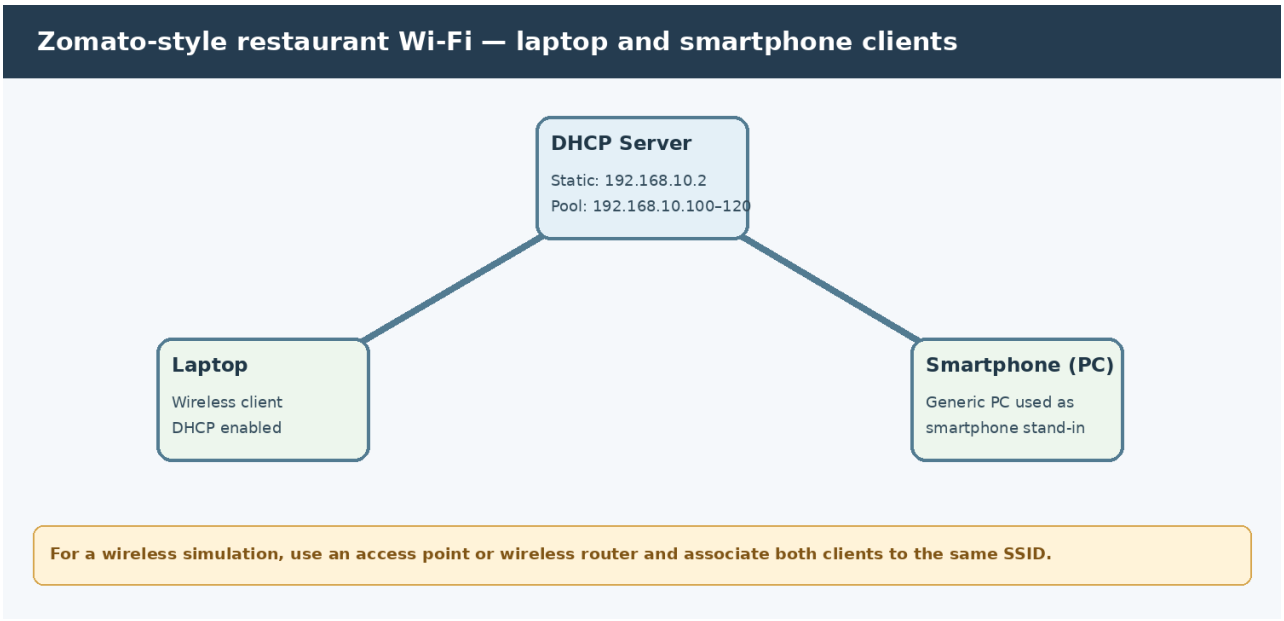


Figure 5. Conceptual restaurant guest network. The generic PC represents a smartphone; this is an illustrative diagram.

13. Create a guest LAN such as 192.168.10.0/24. Configure the DHCP server with a static address such as 192.168.10.2 and subnet mask 255.255.255.0.
14. On Server > Services > DHCP, turn the service On and create a pool named RESTAURANT-GUEST.
15. Set the default gateway to 192.168.10.1, DNS server to 8.8.8.8, Start IP Address to 192.168.10.100, subnet mask to 255.255.255.0 and maximum users to 20.
16. Configure the access point/ wireless router with a guest SSID, for example Restaurant-Guest. Associate the laptop and smartphone-representing device with this wireless network.
17. Set both client devices to obtain their IP configuration automatically using DHCP. Check the assigned address on each device.

4. Restaurant Guest Wi-Fi — Results and Explanation

Example assigned addresses

Illustrative DHCP leases — restaurant guest network

Device	DHCP status	Assigned IPv4 address	Subnet mask
Laptop	DHCP successful	192.168.10.100	255.255.255.0
Smartphone	DHCP successful	192.168.10.101	255.255.255.0

The two clients receive different addresses from the same guest Wi-Fi DHCP pool.

Figure 6. Illustrative DHCP lease results. Replace these example values with the addresses shown in your Packet Tracer simulation.

Client device	Example assigned IP	Subnet mask	Network
Laptop	192.168.10.100	255.255.255.0	Restaurant guest LAN
Smartphone (generic PC)	192.168.10.101	255.255.255.0	Restaurant guest LAN

Why DHCP is useful in a restaurant

- Customers can join the guest Wi-Fi without manually entering an IP address, subnet mask, gateway or DNS server.
- Each connected device can receive a unique address from the pool, allowing multiple customers to use the network at the same time.
- When customers leave and their leases expire or are released, addresses can become available for new guests.
- The restaurant can centrally manage network settings through the DHCP server instead of configuring every guest device individually.

Conclusion

In this simulation, DHCP assigns different IP addresses to the laptop and smartphone-representing PC. This makes guest Wi-Fi easier to use and reduces the work required from restaurant staff and customers.

Final conclusion

The four activities demonstrate how DHCP automatically configures network devices. A DHCP server can allocate addresses to multiple PCs, the DORA sequence explains how a lease is obtained, a one-address pool demonstrates address exhaustion, and a restaurant guest Wi-Fi example shows the practical value of automatic configuration.

Screenshot reminder

The figures in this assignment are illustrative diagrams and sample results, not screenshots captured from Cisco Packet Tracer. For practical submission, add your own Packet Tracer screenshots of the topology, DHCP pool settings, each PC's IP configuration, the one-address failure test and the restaurant guest network.